

Challenges in the Production of a Mixed Reality Theater Dance Performance

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ABSTRACT

Virtual and augmented reality systems are becoming more and more popular in artistic performances. Most of these experiences are based on 360-videos or single-user applications. We produced a mixed reality theater dance performance. An audience of 30 people wore XR-headsets during a theater show. The headsets were networked and shared the same tracking space. Three performers (singer, musician, dancer) performed live and their performance was motion captured and transferred in real-time into the virtual environment. The audience was actively part of the performance and had different possibilities to interact with each other and the environment. This required the integration of different systems and hardware. In this poster, we report our technical setup and discuss the challenges that this entails.

Index Terms: Human-centered computing Human computer interaction (HCI) Interaction paradigms Virtual reality;

1 INTRODUCTION

Recent virtual and augmented reality technology inspires art and culture to invent new kinds of exhibits and performances [1–3]. We have been involved in a theater production of a dance performance that exploited mixed reality (XR) technology to shift boundaries of human experiencing. The production included several challenging aspects that made it unique as far as we know:

- The audience consisted of 30 participants who wore an XR headset during the show.
- All headsets were co-located in the same 10m x 10m large tracking space.
- All headsets were networked and allowed interactions between participants.
- Three performers were performing live in this space: A musician, a singer, and a dancer.
- The dancer wore a motion-capturing suit and this data was transferred in real-time to all headsets.
- The virtual experience was controlled live by an operator (light settings, narrative events, ...).
- Real stage and virtual environment were aligned and the performance included parts with pure real, pure virtual and mixed experiences.

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Figure 1: The stage and set design of the performance included 30 seats. The area was not divided between audience and performers but everyone shared the same space.

2 EXPERIENCE

The theater performance lasted ca. 45 minutes and the audience has worn a headset for ca. 30 minutes. In the beginning, the audience entered the room with the stage setup (see Figure 1) and were seated while the performers already performed. After approximately 10 minutes of performance (music, singing, dancing) in the real world, the audience was asked to mount the headset that was placed on their seat. The participants arrived in a natural virtual environment with some rocks and plants and were able to see a virtual avatar of the dancer. While the performance unfolds, the virtual environment and the virtual avatar changed: Light setting was adjusted, moss and mushrooms grew over the landscape and the avatar grows tentacles (see Figure 2).

Then, the audience was more and more involved into the performance (see Figure 3). First, their own virtual body and their hands have been made visible to themselves. They could start to interact with the environment, e.g. touch corals. In a second step, the avatars of all other participants were made visible to everybody and they could interact with each other now. They were invited to stand up and explore the space. Furthermore, light balls were spawned for each participant and they were given a little task and needed to carry the light balls to a specific point in the environment. At the end of the experience, the passthrough-mode of the headsets was activated and the virtual and real environments were overlaid. Then, the performance ended and the audience unmounted the headsets and left the room.

3 ARCHITECTURE

The central element of our technical setup was an operator software that was used to calibrate and control the setup throughout the whole experience (see Figure 4). Because we needed to enable audience

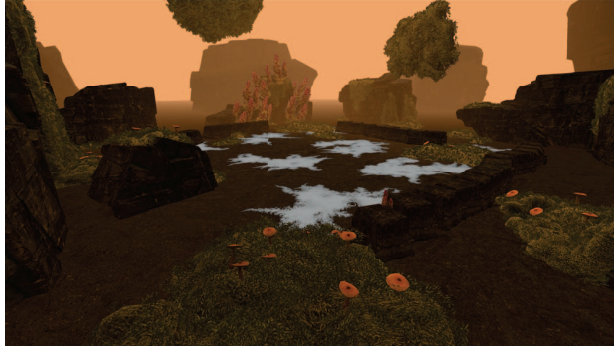


Figure 2: The virtual environment of the experience was aligned to the layout of the real world stage and showed a natural scene with rocks and plants.



Figure 3: The audience and performers during a show. The audience was invited to be actively part of the performance and could explore the whole space.

and performers to free-roam through the space, we decided to use mobile standalone headsets and an untethered motion capturing solution. In this production, we used HTC Vive Focus 3 headsets and a Rokoko 2 suit with a Vive tracking puck. Headsets and suits were networked to the operator software via Wifi.

The headsets and the motion-capturing suit needed to be synchronized in the same shared tracking space. For this, we implemented a calibration mode into the operator software, so it was possible to adjust the position inside the space for each headset and suit individually. For interactions of the participants, handtracking was used.

Furthermore, we were able to provide a spectator screen to a local non-VR audience. We also planned and partly implemented a solution to integrate non-local audience via VR as well as via spectator screen. But due to connection bandwidth issues, we discarded this for the production.

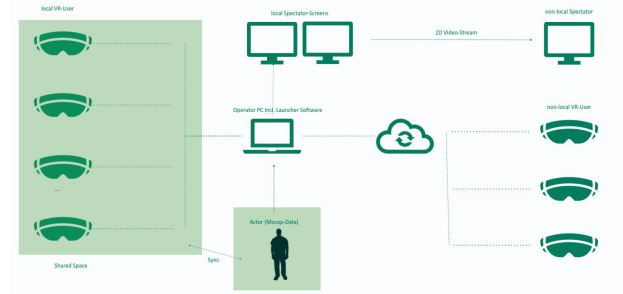


Figure 4: The software and hardware architecture of the technical setup. The non-local audience was planned but was not executed during the production.

4 FEEDBACK & CONCLUSION

So far, six shows of the performance have taken place in a local theater. The audience was pretty mixed with traditional theatergoers with zero XR experience and with young tech-savvy students.

During the development of the software and production of the show, we learned a lot about what we are doing. It was particularly demanding to implement such a variety of different features and integrate so many different hardware components. However, this part worked good. What we did not expect to become a problem is the capability and reliability of the wifi network, even with several modern routers in the room. It worked well in tests but was prone to errors during a show with more than 30 people in the same room. In the later shows, therefore, we limited the audience to 20 persons and it worked much better.

Furthermore, we received valuable feedback from the audience of the shows. The artistic result was praised and a lot of people were surprised during the shows at the diversity of possibilities. Especially, the intersection of artistic and technical approaches was interesting for the audience.

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REFERENCES

- [1] A. Baía Reis and M. Ashmore. From video streaming to virtual reality worlds: an academic, reflective, and creative study on live theatre and performance in the metaverse. *International Journal of Performance Arts and Digital Media*, 18(1):7–28, 2022.
- [2] Y. Kim and H. Lee. Falling in love with virtual reality art: A new perspective on 3d immersive virtual reality for future sustaining art consumption. *International Journal of Human–Computer Interaction*, 38(4):371–382, 2022.
- [3] B. Loveridge. Networked music performance in virtual reality: current perspectives. *Journal of Network Music and Arts*, 2(1):2, 2020.